



Scalability & Future Plan

Date	05 July 2026
Team ID	XXXXXX
Project Name	AI Learning Support Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Interaction history is stored in CSV files instead of a database.	Suitable for small datasets but may become inefficient as the number of interactions grows.	High
2	The application is designed primarily for single-user usage on Streamlit.	Concurrent access by multiple users may affect performance.	Medium
3	AI guidance depends on an active internet connection for the Google Gemini API.	Personalized responses are unavailable when the API or internet connection is not accessible.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports single-user interaction	Deploy the application on a cloud platform

		through a Streamlit application.	with support for multiple concurrent users.
2	Data Storage	Interaction history is maintained in a CSV file.	Replace CSV storage with a relational or cloud database (e.g., MySQL or PostgreSQL) for better scalability.
3	Performance	AI models are cached locally using <code>@st.cache_resource</code> .	Introduce asynchronous processing and optimized inference pipelines for faster response under increased usage.
4	Security	Access is open without user authentication.	Add user authentication, authorization, and secure API key management for production deployment.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the application to a cloud platform for public access.	Short Term	Improve accessibility and allow users to access the application from anywhere.
Phase 2	Replace CSV-based interaction history with a database.	Medium Term	Improve data management, reliability, and scalability.
Phase 3	Introduce user authentication and personalized learning profiles.	Medium Term	Enable secure access and personalized learning experiences.
Phase 4	Enhance analytics with advanced visualizations and personalized learning insights.	Long Term	Provide deeper analysis of user learning patterns and progress.

Summary

The current implementation successfully meets the project's objectives by providing:

- Emotion prediction using **BiLSTM** and **BERT**
- Mixed emotion analysis
- Personalized learning guidance using the **Google Gemini API**
- Previous interaction history
- Analytics dashboard with Plotly visualizations
- Responsive Streamlit user interface

To support larger-scale deployments, future work may focus on cloud deployment, database integration, enhanced security, and advanced analytics. These enhancements are proposed as future improvements and are **not part of the current implementation**, ensuring the documentation remains accurate and consistent with the project you built.